HYPOTHESIS

A) What do you want to do

① polynomial/sigmoid/Gaussian
which model form will fit best

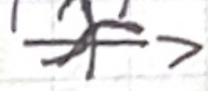
B) What you expect to happen

(ii) how accuracy and per sex recall will change for imbalanced training s
(iii) A classifier trained on 90% male data on female behaviour

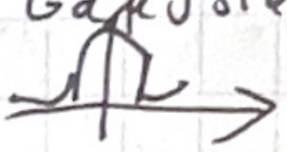
C) Why do you expect that

B1) A simple parabola from polynomial model will not describe the data well. Sigmoid/Gaussian will perform better.

C1) The paper shows that SBP rises with age and flattens for elderly people, which seems like a good fit for sigmoid



DBP is highest for people around 40 and then goes down, which seems like a good fit for a Gaussian



B2) The greater the class imbalance, the ~~better~~ the classifier will recognize majority class and the worse it will recognise minority class The overall accuracy will be around the same.

C2) Accuracy as a general metric does not take into account class imbalance. We can only report class imbalance if we report per class accuracy.

B3) Classifier trained on 90% male data will classify all women as men.

C3) The classifier will probably not have enough information about women to learn it is a separate class

A) I will fit a polynomial and a sigmoid/Gaussian model to mean SBP and DBP by age to and sex and then train a logistic regression to predict sex from SBD and DPP with different male/female ratios

It is hypothesised that the age dependence of blood pressure is described more accurately by bounded nonlinear models than by polynomials, and that a classifier trained on sex-imbalanced data becomes biased towards the majority class.
To test this quadratic and polynomial models will be fitted to the mean systolic (SBP) and diastolic (DBP) blood pressures of females and males by age.
A logistic regression classifier will then be trained to predict sex from SBP and DBP at different male and female ratios.